APPLIED PHYSICS LAB

I B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A5BS15	BSC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100
Contact Classes: 0	**Tutorial Classes: 0**	**Practical Classes: 39**				**Total Classes: 39**		

COURSE OBJECTIVES

The course should enable the students to:

1. To provide an experimental foundation for the theoretical concepts introduced in the lectures
2. To teach how to make careful experimental observations and how to think about and draw conclusions from such data
3. To help students understand the role of direct observation in physics and to distinguish between inferences based on theory and the outcomes of experiments
4. To introduce the concepts and techniques which have a wide application in experimental science but have not been introduced in the standard courses
5. To teach how to write a technical report this communicates scientific information in a clear and concise manner

COURSE OUTCOMES

By the end of the course students will be able:

1. **Analyze** the electric properties of semiconductor materials by determining energy gap of semiconductors, threshold voltage of LEDs and efficiency issues of solar cell with careful experimental and draw conclusions from such data
2. **Evaluate** the mechanical properties of a given material using dynamic method in torsional pendulum
3. **Estimate** the optical properties of light such as diffraction by using grating material for calculation of the wavelength of Laser, and to determine acceptance angle, NA of optical fiber using OFC and determine the value off plank's constant
4. **Analyze** the electromagnetic properties by using Stewart Gee's experiment and determining the Quality factor, resonance frequency of a given LCR circuit

LIST OF EXPERIMENTS

Experiment - 1	**Energy gap of P-N junction diode:** To determine the energy gap of a semiconductor diode
Experiment - 2	**Solar Cell:** To study the V-I and P-I characteristics of solar cell
Experiment - 3	**Light Emitting Diode:** Plot V-I characteristics of light emitting diode Plot V-I characteristics of light emitting diode
Experiment - 4	**Hall effect:** To determine Hall co-efficient of a given semiconductor
Experiment - 5	**PIN Photo Diode** To study the V-I Characteristics of Photo Diode by calculating the photo current.
Experiment - 6	**Optical fiber:** To determine the numerical aperture and acceptance angle of an optical fiber
Experiment - 7	**LASER:** To determine the wavelength of a given laser source by using diffraction grating method

Experiment - 8	**LCR Circuit:** To determine the Resonance frequency and Quality factor of a LCR Circuit
Experiment - 9	**Thermistor:** To study the variation of resistance with respect to temperature using thermistor.
Experiment - 10	**Torsional Pendulum:** To determine the rigidity modulus of a given metal wire by using Torsional pendulum
Experiment - 11	**Plank's Constant:** To determine value of plank's constant using by measuring radiation in fixed spectral range
Experiment - 12	**Stewart Gee's experiment:** To study the variation of magnetic field along the axis of a circular coil

REFERENCE BOOKS

1. "Semiconductor Physics and Devices: Basic Principles" by Donald A Neamen
2. "Optics, Principles and Applications" by K K Sharma.
3. "Principles of Optics" by M Born and E Wolf.
4. "Oscillations and Waves" by Satya Prakash and Vinay Dua
5. "Waves and Oscillations" by N Subrahmanyam and Brij Lal

WEB REFERENCES

1. **http://www.arxiv.org/pdf/1510.00032**
2. http://www nptel.ac.in/courses/122103010/
3. **http://www.** researchgate.net/.../276417736_Video_Presentations_in_Engineering-Ph...
4. **http://www.** wileyindia.com/engineering-physics-theory-and-practical.html